

Task #322 v0.9 — A_sub Spin(5) cover formal incorporation: per-generator cover classification for super-quiver structure

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A_sub v0.9 — Spin(5) cover formal incorporation

1. Cal #29 STANDING question-shape audit (applied at design)

Question: “Across A_sub’s 15 generators (14 from v0.1 + σ_{BF} explicit), which require Spin(5) cover representation when acting on the fermion sublattice (half-integer K-types) vs operate with SO(5) tensor representation alone on the boson sublattice (integer K-types)?”

Audit: - Structurally determined? YES — standard representation theory of Spin(5) $\subset$ Sp(2) vs SO(5); cover requirement determined by half-integer-spin content - Back-fittable? NO — cover classification follows from generator action on Wallach K-types - Pre-suppositions? Step 10 SVC CANDIDATE (spin algebra requires Spin(5) cover on fermion sublattice) + Cal #132 SVC for steps 1-8 + σ_{BF} vs γ^5 disambiguation

Pass: structural question; standard representation theory.

2. Step 10 finding recap (Spin(5) cover required on fermion sublattice)

Per Lyra_A_sub_Commutator_S_i_S_j_Across_Sublattices_v0_1.md (yesterday SVC CANDIDATE):

$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk} \hat{S}_k$ universally on $H^2(D_{IV}^5)$, with REPRESENTATION DIFFERENCE across σ_{BF} Z_2 sublattices:

Sublattice	σ_{BF}	SO(5) representation	Cover requirement
Boson (integer m_1)	+1	Tensor reps; integer spin	SO(5) sufficient

Sublattice	σ_{BF}	SO(5) representation	Cover requirement
Fermion (half-integer m_1)	-1	Spinor reps; half-integer spin	Spin(5) cover REQUIRED

$\text{Spin}(5) \cong \text{Sp}(2)$ (compact symplectic) is the universal double cover of $\text{SO}(5)$. For fermion K-types, half-integer-spin representations are projective representations of $\text{SO}(5)$ — they require lift to $\text{Spin}(5)$ for unambiguous action.

3. K-cover structure on $H^2(D_{\text{IV}}^5)$

The full K-cover structure for substrate's bounded symmetric domain $D_{\text{IV}}^5 = \text{SO}_0(5,2)/[\text{SO}(5) \times \text{SO}(2)]$:

$K_{\text{cover}} = \text{Spin}(5) \times \text{Pin}(2)$ (universal cover of $K = \text{SO}(5) \times \text{SO}(2)$)

Components: - $\text{Spin}(5) \cong \text{Sp}(2)$: double cover of $\text{SO}(5)$; acts on m_1 component (S^4 side, Hua coordinates) - $\text{Pin}(2)$: double cover of $\text{SO}(2)$; acts on m_2 component (S^1 side, Möbius/Bergman phase) - $\sigma_{\text{BF}} Z_2$ grading: distinguishes integer- m K-types (boson sublattice, lifts trivially) from half-integer- m K-types (fermion sublattice, requires non-trivial cover lift) - γ^5 Dirac chirality: Z_2 acting WITHIN fermion sublattice; eigenvalues ± 1 for L/R Weyl spinors

The universal cover structure is $\text{Spin}(5) \times \text{Pin}(2)$ with the $\sigma_{\text{BF}} Z_2$ grading distinguishing how it lifts at each K-type.

4. Per-generator $\text{Spin}(5)$ cover classification

A_sub's 15 generators (v0.1 14 + σ_{BF} explicit per disambiguation cleanup) classified by $\text{Spin}(5)$ cover requirement:

Class A — Diagonal (no cover issue)

Act diagonally on K-types; eigenvalues independent of cover:

Generator	Action	Cover required?
$\hat{H}_{\text{sub}}$	$C_2(K) \cdot V_K$	No
$\hat{N}$	$(m_1 + m_2) \cdot V_K$	No
σ_{BF}	$\varepsilon_K \cdot V_K$ (Z_2 grading)	No
$\hat{\gamma}^5$	$\chi_K \cdot V_K$ on fermion sublattice (undefined on bosons)	Yes (defined only via $\text{Spin}(5) \supset \text{Spin}(3)$ chirality structure)
$\hat{Q}$	$m_2 \cdot V_K$	No
$\hat{I}_3$	$T_3^{\text{weak}} \text{ eigenvalue} \cdot$ V_K (Cartan)	No

5 of 6 diagonal generators require no cover lift; $\hat{\gamma}^5$ as Dirac chirality requires Spin(5) cover for its definition on fermion sublattice.

Class B — m_1 -direction action (SO(5) tensor / Spin(5) spinor)

Raise/lower m_1 within K-type multiplet:

Generator	Action	Cover required?
$\hat{x}_i$ (Hua coords, $i=1..n_C=5$)	$(m_1, m_2) \rightarrow (m_1 \pm 1, m_2)$	Yes on fermion sublattice
$\hat{p}_i$ (Wirtinger derivative, $i=1..5$)	$(m_1, m_2) \rightarrow (m_1 \pm 1, m_2)$	Yes on fermion sublattice
$\hat{L}_i$ (so(5) generators, 10 total)	Within SO(5) multiplet	Yes on fermion sublattice

On boson sublattice (integer m_1): SO(5) tensor representations suffice. On fermion sublattice (half-integer m_1): require Spin(5) lift for proper Wallach K-type action on half-integer-spin representations.

Class C — Spinor algebra (Spin(5) cover-required)

Per step 10 SVC CANDIDATE:

Generator	Action	Cover required?
$\hat{S}_i$ (3 spin generators)	SU(2)-like spin algebra	Yes on fermion sublattice (step 10)

Spin algebra requires Spin(5) cover for half-integer-spin representations.

Class D — Discrete symmetries (Pin(5) extension on fermion sublattice)

Discrete symmetries lift via Pin(5) = double cover of O(5):

Generator	Action	Cover required?
$\hat{T}$ (time reversal, anti-unitary)	$V_K \rightarrow \bar{V}_K$ with phase	Pin(5) lift on fermion sublattice
$\hat{C}$ (charge conjugation)	$V_K \rightarrow V_{\bar{K}}$ with phase	Pin(5) lift on fermion sublattice
$\hat{P}_{op} = \hat{\gamma}^5 \circ \sigma$ (Möbius + Pin(2) Z_2 lift)	$V_K \rightarrow V_{\bar{K}}$ with $\hat{\gamma}^5$ phase	Built-in Spin(5) $\times$ Pin(2) lift via $\hat{\gamma}^5$ factor

Discrete symmetries naturally extend to Pin(5) = Spin(5) $\cup$ (reflection $\times$ Spin(5)) acting on universal cover.

Class E — Bell-CHSH (rank-1 projector at vacuum)

Acts within boson ground state $V_{(0,0)}$; rank-1 projector:

Generator	Action	Cover required?
$\hat{B} = \beta \cdot$	$V_{(0,0)} \rangle \langle V_{(0,0)}$	

Bell-CHSH at substrate vacuum (boson; integer K-type) requires no cover lift for its rank-1 projector form. For fermion-bound states (e.g., entangled spinor pairs), the Bell-CHSH operator must lift to $\text{Spin}(5)$ cover for proper action.

Class F — Substrate-tick (lifts naturally)

Generator	Action	Cover required?
$\hat{T}_{\text{tick}}$ (substrate-tick transition)	$V_K \rightarrow V_{\{K+1\}}$ per Koons tick (model-dep)	Naturally lifts via cover structure

$\hat{T}_{\text{tick}}$ is a discrete unitary acting on $H^2(D_{IV}^5)$; preserves $\sigma_{BF} Z_2$ grading per cyclic structure; lifts naturally to $\text{Spin}(5) \times \text{Pin}(2)$ cover.

Class G — Gauge fiber (transverse to spinor structure)

Generator	Action	Cover required?
$\hat{C}_3$ (color $\text{SU}(3)$, 8 generators)	Acts on color triplet fiber	No (transverse)
$\hat{I}_3$ raising/lowering (weak $\text{SU}(2)$)	Acts on weak doublet fiber	No (transverse)

Gauge fiber operators are transverse to the spinor cover structure (color + weak isospin act in gauge fiber, not on Wallach K-type spin content). No cover lift needed.

5. Universal cover structure summary

A_{sub} on Bergman $H^2(D_{IV}^5)$ has:

- Underlying group: $K = \text{SO}(5) \times \text{SO}(2)$
- Universal cover: $K_{\text{cover}} = \text{Spin}(5) \times \text{Pin}(2)$
- Boson sublattice: integer K-types $V_{(m_1, m_2)}$; $\text{SO}(5) \times \text{SO}(2)$ representations suffice; no non-trivial cover lift
- Fermion sublattice: half-integer K-types $V_{(m_1, m_2)}$; $\text{Spin}(5) \times \text{Pin}(2)$ cover representations REQUIRED

- Cross-sublattice operators: structurally zero (NO SUSY per Five-Absence Principle)
- $\sigma_{BF} Z_2$ grading: distinguishes the two sublattices universally
- γ^5 Dirac chirality: Z_2 acting WITHIN fermion sublattice via $Spin(5)$ cover structure
- $Pin(2) Z_2$: discrete reflection on $Pin(2)$ side for parity operations

The substrate has explicit Z_2 -graded super-algebra structure via $\sigma_{BF} Z_2$ grading + cover requirement.

6. Z_2 -graded super-quiver structure (multi-phase quiver v0.2+ pre-requisite)

A_{sub} 's per-generator cover classification gives the substrate quiver Q a Z_2 -graded super-quiver structure:

6.1 Node classification

- Boson nodes (Q_+): integer K-type vertices; $\sigma_{BF} = +1$; $SO(5) \times SO(2)$ representation
- Fermion nodes (Q_-): half-integer K-type vertices; $\sigma_{BF} = -1$; $Spin(5) \times Pin(2)$ cover representation

6.2 Arrow classification

- Even arrows (degree-0): preserve σ_{BF} sublattice; bosonic to bosonic OR fermionic to fermionic
- Odd arrows (degree-1): would flip σ_{BF} sublattice; STRUCTURALLY ABSENT per NO SUSY (Five-Absence Principle)

The substrate quiver is a trivial Z_2 -graded super-quiver with only even arrows (no degree-1 super-symmetry transitions).

6.3 Implication for Hall algebra

Per Kapranov + Schiffmann + Ringel-Green super-Hall theory: super-quiver Hall algebra construction over Z_2 -graded categories requires: - Z_2 -graded vector space representations at each node - Even-degree linear maps for each arrow - Z_2 -grading consistency for path composition

Substrate's super-quiver IS in this framework: - $V_x = SO(5) \times SO(2)$ representation at boson nodes; $Spin(5) \times Pin(2)$ representation at fermion nodes - All arrows degree-0 (no SUSY) - Z_2 grading via σ_{BF} preserved by all arrows

Hall algebra construction (Ringel-Green) applies cleanly to substrate's Z_2 -graded super-quiver.

7. Multi-phase quiver v0.2+ readiness

With A_{sub} v0.9 $\text{Spin}(5)$ cover formal incorporation complete, the multi-phase quiver v0.2+ Hall algebra work has rigorous foundation:

- Quiver Q explicitly defined (Lyra v0.8 + v0.9 super-grading)
- Path algebra kQ can be constructed over field k (substrate-natural $k = \mathbb{C}$ for Bergman, or $k = \text{GF}(2^X)$ at finite-field substructure)
- Relations R from Cal #132 SVC commutators (with σ_{BF} vs γ^5 disambiguation applied per v0.7 cleanup)
- Super-algebra structure explicit (Z_2 -graded; no SUSY arrows; clean super-Hall construction)
- Cover structure explicit (boson nodes $\text{SO}(5) \times \text{SO}(2)$; fermion nodes $\text{Spin}(5) \times \text{Pin}(2)$)

This is the prerequisite Casey approved this morning's sequencing for. v0.2+ Hall algebra explicit framework starts next pull.

8. Honest scope (Cal #27 STANDING + Cal #29 STANDING)

What's RATIFIED/SVC: - Step 10 $[\hat{S}_i, \hat{S}_j]$ SVC CANDIDATE finding ($\text{Spin}(5)$ cover required on fermion sublattice) - Standard representation theory of $\text{Spin}(5) \cong \text{Sp}(2)$ as universal double cover of $\text{SO}(5)$ - $\text{Pin}(5)$ extension for discrete symmetries (standard) - σ_{BF} vs γ^5 disambiguation (v0.7 cleanup applied) - $\text{Pin}(2)$ Z_2 grading on Wallach K-types per T2471 RATIFIED

What's FRAMEWORK in v0.9: - Per-generator cover classification table (Section 4) - Z_2 -graded super-quiver structural reading (Section 6) - Hall algebra readiness check (Section 7)

What's INTERPRETIVE in v0.9: - "Substrate quiver is trivial Z_2 -graded super-quiver (no degree-1 SUSY arrows)" — consistent with Five-Absence Principle (RATIFIED Casey-named) but requires explicit verification that NO degree-1 arrows arise from substrate-mechanism

What's NOT in v0.9 (multi-phase quiver v0.2+ work): - Explicit Ringel-Green Hall algebra construction on substrate's super-quiver - Kac-Moody Lie algebra identification of A_{sub} via Hall-algebra construction - Macdonald-like 5-parameter deformation (last night's Hall conversation) - α -as-Hall-Littlewood-at- N_{max} -level formalization

Cal #27 STANDING reflexive trigger count: 1 trigger (Section 6 super-quiver reading feels structurally elegant). Honest scope check — standard super-quiver theory + step 10 SVC CANDIDATE + Five-Absence Principle (RATIFIED); no new substantive claim beyond clean formalization.

Cal #29 STANDING pass: per-generator classification is structurally determined; not back-fittable; standard representation theory.

9. Coordination

Cal: Thread 4 typing on Section 6 super-quiver classification is Type B algebraic (standard super-Hall framework); no Type C level-crossing claim in v0.9 (house-keeping doc).

Elie: standing reactive; v0.9 doesn't add new toy candidates.

Grace: catalog entries for $\text{Spin}(5) \times \text{Pin}(2)$ cover structure across A_{sub} generators; relevant for SPLP Phase 2 audit at substrate-algebraic level.

Keeper: v0.9 completes A_{sub} v0.1 $\rightarrow$ v0.9 algebraic foundation; multi-phase quiver v0.2+ Hall algebra work begins with rigorous super-quiver prerequisite in place.

— Lyra, A_{sub} v0.9 $\text{Spin}(5)$ cover formal incorporation v0.1 filed Wednesday 2026-05-27 ~09:00 EDT per Casey approval of sequencing. FRAMEWORK + HOUSEKEEPING grade. Step 10 SVC CANDIDATE algebraic content formalized across A_{sub} generator list; universal cover structure $\text{Spin}(5) \times \text{Pin}(2)$ explicit; super-quiver Z_2 -graded structure ready for multi-phase quiver v0.2+ Hall algebra construction (next pull). Standard representation theory throughout; no new substantive claims; clean prerequisite hygiene.